

Question_4:

Code:

```
1 // Byteland has n cities and m roads. Add the minimum number of roads so that
2 // every city becomes connected, and print the new roads that must be added.
3 #include <iostream>
4 using namespace std;
5
6 #define MAXN 100005
7
8 int parent[MAXN];
9 int find(int i)
10 {
11     if (parent[i] == i)
12         return i;
13     return parent[i] = find(parent[i]);
14 }
15 void unite(int i, int j)
16 {
17     int rootI = find(i);
18     int rootJ = find(j);
19     if (rootI != rootJ)
20         parent[rootI] = rootJ;
21 }
22 int main()
23 {
24     int n, m;
25     cin >> n >> m;
26     for (int i = 1; i <= n; i++)
27         parent[i] = i;
28
29     while (m--)
30     {
31         int u, v;
```

```
32     cin >> u >> v;  
33     unite(u, v);  
34 }  
35 int components[MAXN], count = 0;  
36  
37 for (int i = 1; i <= n; i++)  
38     if (find(i) == i)  
39         components[count++] = i;  
40  
41 cout << count - 1 << endl;  
42  
43 for (int i = 0; i < count - 1; i++)  
44     cout << components[i] << " " << components[i + 1] << endl;  
45  
46 return 0;  
47 }
```

Output:

Test Case_1:

```
4 2  
1 2  
3 4  
1  
2 4  
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
4 2  
2 3  
2 4  
1  
1 4  
PS C:\Users\Prabin Yadav\
```